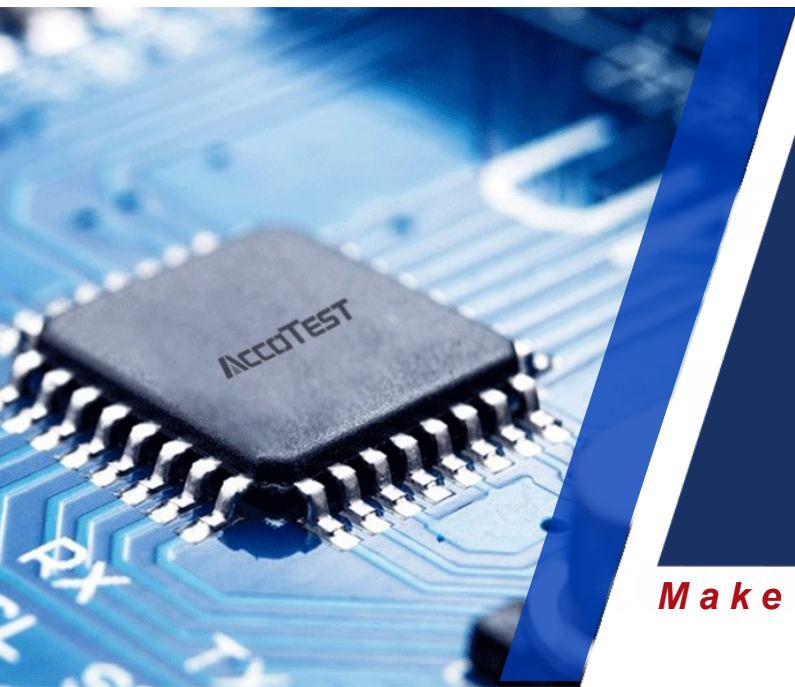




AccoTEST STS8300 应用培训

QVMe 单板硬件及应用

2022/2/18

Make Testing More Valuable

目录

- 单板特性
- 指标
- 原理框图及外观结构
- 失真度测量特性曲线
- 槽位及对应PogoPin定义
- QVMe编程介绍

• 单板特性

SM8212 四通道全浮动差分电压表模块是 STS 8300 系统测试头内的标准模块，模块提供四路全浮动差分电压表。其主要性能如下：

- 系统最多支持 4 块 QVMe 模块工作
- 每模块四通道，每通道独立浮动
- 具备最大 1MSPS 18bit 高精度测量模式
- 具备最大 10MSPS 12bit 高速测量模式
- 高精度模式 10 个量程：
- $\pm 100V$ ， $\pm 50V$ ， $\pm 20V$ ， $\pm 10V$ ， $\pm 5V$ ， $\pm 2V$ ， $\pm 1V$ ， $\pm 500mV$ ， $\pm 200mV$ ， $\pm 100mV$
- 高速模式 3 个量程：
- $\pm 4V$ ， $\pm 2V$ ， $\pm 1V$
- 采样速率可调
- 每通道具备 64K*18bit 的存储器
- 具备板上硬件数据处理功能，包括：
- MAX、 MIN、 AVG、 Trigger Point、 RMS、 FFT、 THD、 SNR、 SINAD
- 隔离电压 500V
- 支持多工位并行测试工作模式
- 本板保存校准数据

• 指标

Sample Depth	64K/channel	
Measurement Type	MAX、MIN、AVG、Trigger point、RMS、FFT、THD、SNR、SINAD	
High Precision Mode	Measurement Range	$\pm 100V$, $\pm 50V$, $\pm 20V$, $\pm 10V$, $\pm 5V$, $\pm 2V$, $\pm 1V$, $\pm 500mV$, $\pm 200mV$, $\pm 100mV$
	Impedance	$>1G$ ($\leq \pm 10V$ Range) $1M$ ($> \pm 10V$ Range)
	Maximum Sample Rate	1MSPS
	Range of Sampling Interval	1us~1300us
	Resolution of Sampling Interval	0.1us
	Resolution	18bit
	LPF	10KHz、40KHz、100KHz、400KHz
	THD	-90dBTyp. ⁽¹⁾ (Vpp=10.0V, Freq=1KHz,LPF=40KHz)
	SNR	78dBTyp. ⁽²⁾ (Vpp=10.0V, Freq=1KHz,LPF=40KHz)
	SINAD	76dBTyp. ⁽²⁾ (Vpp=10.0V, Freq=1KHz,LPF=40KHz)

- 指标

High Speed Mode	Measurement Range	$\pm 4V, \pm 2V, \pm 1V$
	Impedance	1M
	Maximum Sample Rate	10MSPS
	Range of Sampling Interval	0.1us~1.0us
	Resolution of Sampling Interval	0.1us
	Resolution	12bit
	Bandwidth	>10MHz
	THD	-78dBTyp. ⁽¹⁾ (Vpp=4.0V, Freq=100KHz)

(1) 2nd to 19th harmonic

(2) DC to 19KHz frequency band

• 指标

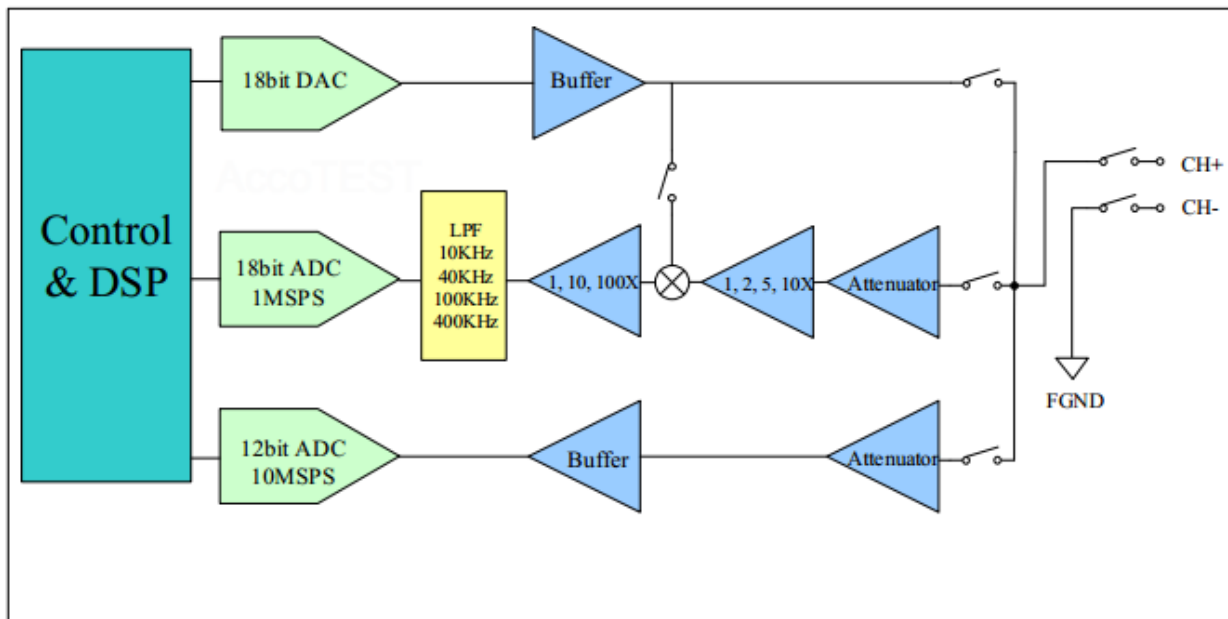
High Precision Mode

Range	Input Z	Resolution	Accuracy{ $\pm$ (Offset+ $\%$ Rdg)}
$\pm 100\text{V}$	1M	781.3uV	$\pm(10\text{mV}+0.01\%)$
$\pm 50\text{V}$		390.6uV	$\pm(5\text{mV}+0.01\%)$
$\pm 20\text{V}$		156.3uV	$\pm(2\text{mV}+0.01\%)$
$\pm 10\text{V}$	>1G	78.1uV	$\pm(1\text{mV}+0.01\%)$
$\pm 5\text{V}$		39.1uV	$\pm(500\text{uV}+0.01\%)$
$\pm 2\text{V}$		15.6uV	$\pm(200\text{uV}+0.01\%)$
$\pm 1\text{V}$		7.8uV	$\pm(150\text{uV}+0.01\%)$
$\pm 500\text{mV}$		3.9uV	$\pm(100\text{uV}+0.01\%)$
$\pm 200\text{mV}$		1.56uV	$\pm(100\text{uV}+0.01\%)$
$\pm 100\text{mV}$		0.78uV	$\pm(100\text{uV}+0.01\%)$

High Speed Mode

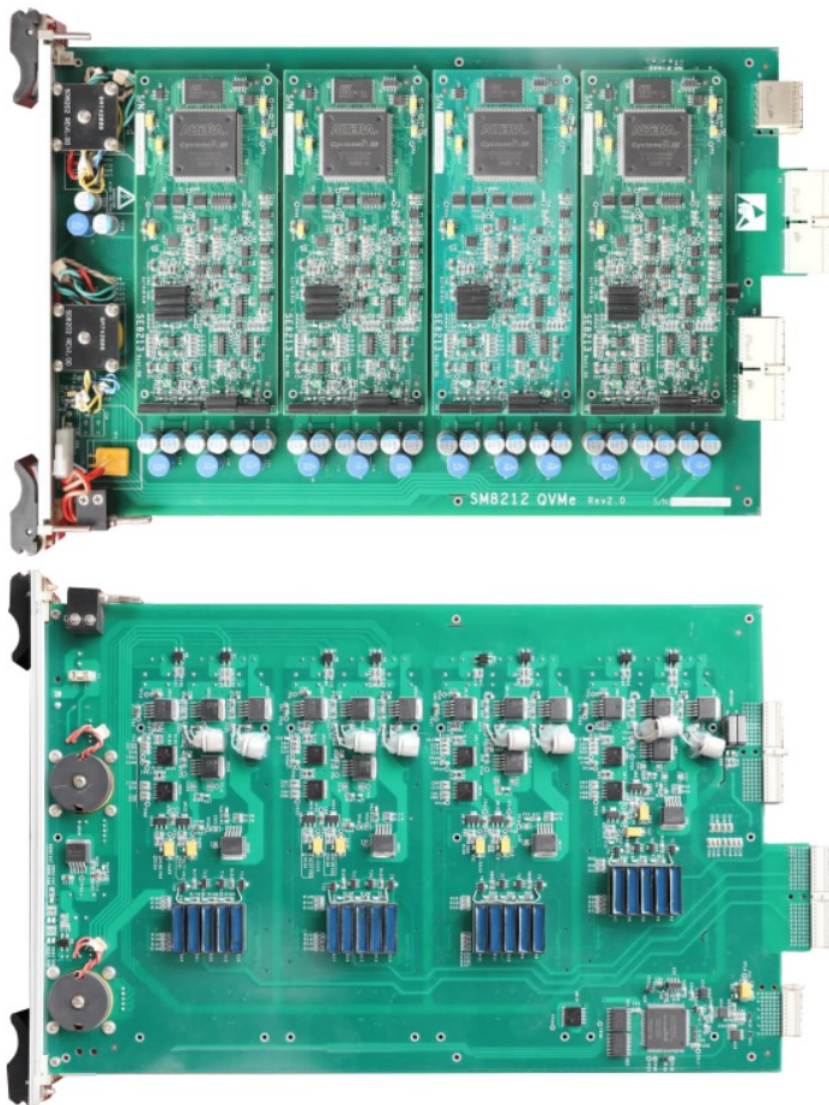
Range	Input Z	Resolution	Accuracy{ $\pm$ (Offset+ $\%$ Rdg)}
$\pm 4\text{V}$	1M	2.44mV	$\pm(4\text{mV}+0.1\%)$
$\pm 2\text{V}$		1.22mV	$\pm(2\text{mV}+0.1\%)$
$\pm 1\text{V}$		0.61mV	$\pm(1\text{mV}+0.1\%)$

• 原理框图



QVMe Block Diagram

- 外观结构



失真度测量特性曲线

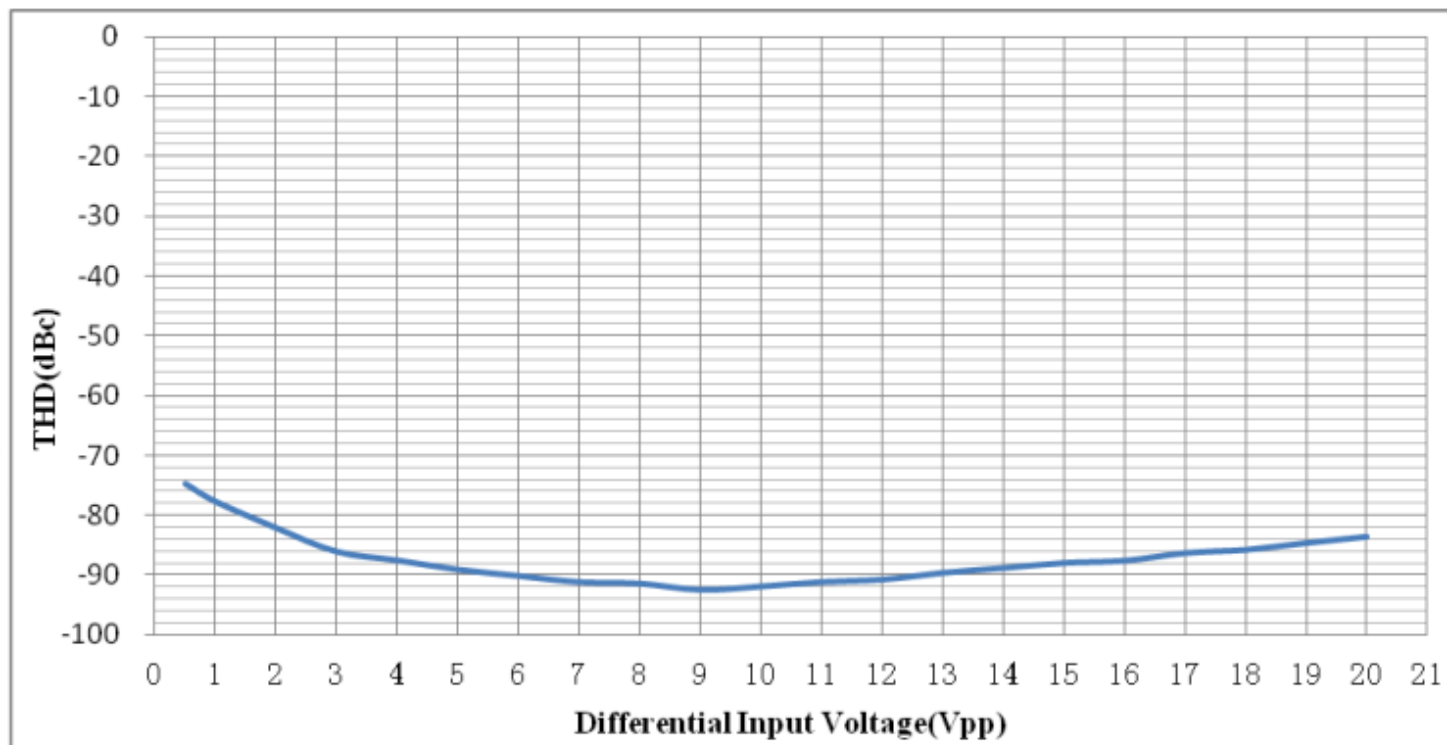


Fig 2- 20 18bits 1MSPS, 10V Range, 400K Filter, 2nd to 19th harmonic @0.9765625KHz Sine

失真度测量特性曲线

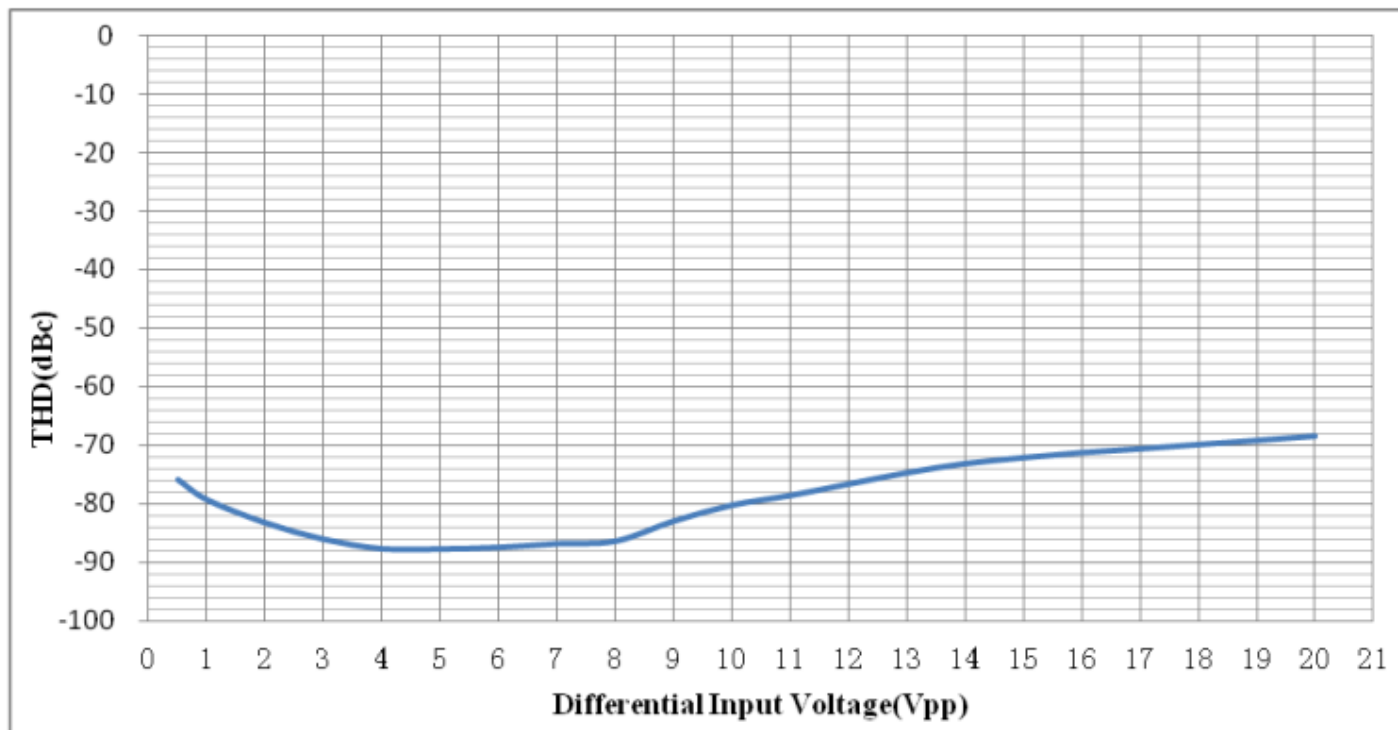


Fig2- 21 18bits 1MSPS, 10V Range, 400K Filter, 2nd to 19th harmonic@9.765625KHz Sine

失真度测量特性曲线

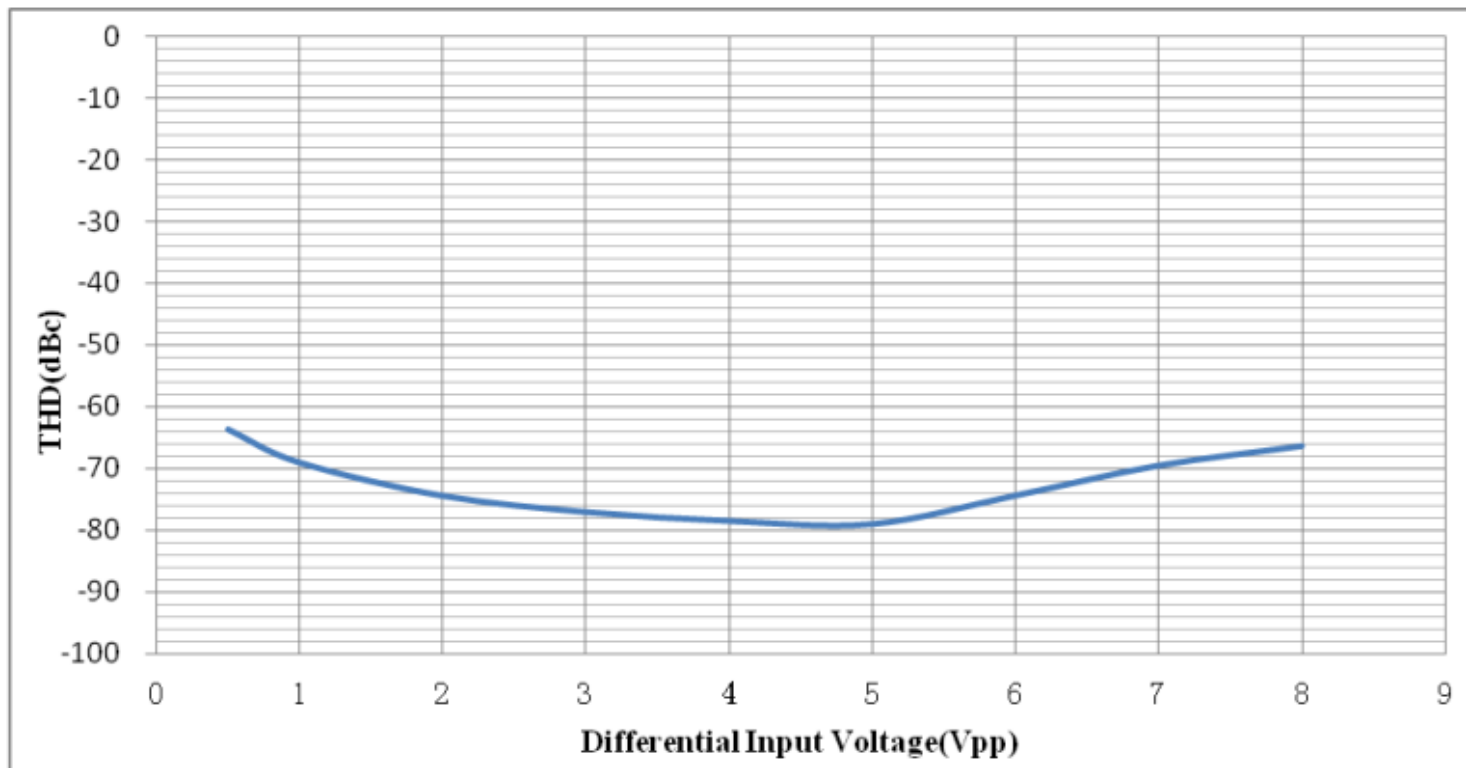


Fig2- 22 12bits 10MSPS, 4V Range, 2nd to 19th harmonic @97.65625KHz Sine

槽位及对应PogoPin定义

QVMe对应的Analog槽位（TIB类型相关）

33	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	34
NULL	Analog	Analog	Analog	Analog	Analog	Analog	Analog	Analog	DCM	QTMUe	Analog	Analog	Analog	Analog	Analog	Analog	CBITe
CBITe	Analog	Analog	Analog	Analog	Analog	Analog	Analog	Analog	DCM	QTMUe	Analog	Analog	Analog	Analog	Analog	Analog	Cal
36	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	35

A28D4 Config

33	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	34
NULL	Analog	Analog	Analog	Analog	Analog	Analog	TMU	Digital	Digital	TMU	Analog	Analog	Analog	Analog	Analog	Analog	CBITe
CBITe	Analog	Analog	Analog	Analog	Analog	Analog	TMU	Digital	Digital	TMU	Analog	Analog	Analog	Analog	Analog	Analog	CAL
36	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	35

A24D8 Config

槽位及对应PogoPin定义

From PogoPin Tool

P101_slot5slot6

	AGND	G_11 FANOUT	SH_11 FANOUT	FH_11 FANOUT	SL_6-11	FL_6-11	FL_6-11	G_10 FANOUT	SH_10 FANOUT	FH_10 FANOUT	
	AGND	G_11	SH_11	FH_11	SL_6-11	FL_6-11	FL_6-11	G_10	SH_10	FH_10	
G_9	SH_9	FH_9	G_8	SH_8	FH_8	G_7	SH_7	FH_7	G_6	SH_6	FH_6
					CH_3-	G_5 FANOUT	SH_5 FANOUT	FH_5 FANOUT	G_3	SH_3	FH_3
					CH_3+	G_5	SH_5	FH_5	G_2	SH_2	FH_2
						SL_0-5	FL_0-5	FL_0-5	SL_0-5	FL_0-5	FL_0-5
						G_4 FANOUT	SH_4 FANOUT	FH_4 FANOUT	G_1	SH_1	FH_1
						G_4	SH_4	FH_4	G_0	SH_0	FH_0
		CH_2-			CH_2+			CH_1-			CH_1+
	AGND									CH_0-	
	AGND									CH_0+	
12	11	10	9	8	7	6	5	4	3	2	1

K
J
I
H
G
F
E
D
C
B
A

Open Config.slot

Resource:

QWfe

Slot:

slot5

Serial NO:

0

☒ AutoSwitch

Config File Path:

C:/AccoTest/AccoTEST System/ConfigEx.slot

Cell Channel And Slot:

P000: P001: P002: P003: P004: P005: P006: P007: P008: P009: P010: P011: P012: P013: P014: P015: P016: P017: P018: P019: P020: P021: P022: P023: P024: P025: P026: P027: P028: P029: P030: P031: P032: P033: P034: P035: P036: P037: P038: P039: P040: P041: P042: P043: P044: P045: P046: P047: P048: P049: P050: P051: P052: P053: P054: P055: P056: P057: P058: P059: P060: P061: P062: P063: P064: P065: P066: P067: P068: P069: P070: P071: P072: P073: P074: P075: P076: P077: P078: P079: P080: P081: P082: P083: P084: P085: P086: P087: P088: P089: P090: P091: P092: P093: P094: P095: P096: P097: P098: P099: P100: P101: P102: P103: P104: P105: P106: P107: P108: P109: P110: P111: P112: P113: P114: P115: P116: P117: P118: P119: P120: P121: P122: P123: P124: P125: P126: P127: P128: P129: P130: P131: P132: P133: P134: P135: P136: P137: P138: P139: P140: P141: P142: P143: P144: P145: P146: P147: P148: P149: P150: P151: P152: P153: P154: P155: P156: P157: P158: P159: P160: P161: P162: P163: P164: P165: 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P1282: P1283: P1284: P1285: P1286: P1287: P1288: P1289: P1290: P1291: P1292: P1293: P1294: P1295: P1296: P1297: P1298: P1299: P1300: P1301: P1302: P1303: P1304: P1305: P1306: P1307: P1308: P1309: P1310: P1311: P1312: P1313: P1314: P1315: P1316: P1317: P1318: P1319: P1320: P1321: P1322: P1323: P1324: P1325: P1326: P1327: P1328: P1329: P1330: P1331: P1332: P1333: P1334: P1335: P1336: P1337: P1338: P1339: P1340: P1341: P1342: P1343: P1344: P1345: P1346: P1347: P1348: P1349: P1350: P1351: P1352: P1353: P1354: P1355: P1356: P1357: P1358: P1359: P1360: P1361: P1362: P1363: P1364: P1365: P1366: P1367: P1368: P1369: P1370: P1371: P1372: P1373: P1374: P1375: P1376: P1377: P1378: P1379: P1380: P1381: P1382: P1383: P1384: P1385: P1386: P1387: P1388: P1389: P1390: P1391: P1392: P1393: P1394: P1395: P1396: P1397: P1398: P1399: P1400: P1401: P1402: P1403: P1404: P1405: P1406: P1407: P1408: P1409: P1410: P1411: P1412: P1413: P1414: P1415: P1416: P1417: P1418: P1419: P1420: P1421: P1422: P1423: P1424: P1425: P1426: P1427: P1428: P1429: P1430: P1431: P1432: P1433: P1434: P1435: P1436: P1437: P1438: P1439: P1440: P1441: P1442: P1443: P1444: P1445: P1446: P1447: P1448: P1449: P1450: P1451: P1452: P1453: P1454: P1455: P1456: P1457: P1458: P1459: P1460: P1461: P1462: P1463: P1464: P1465: P1466: P1467: P1468: P1469: P1470: P1471: P1472: P1473: P1474: P1475: P1476: P1477: P1478: P1479: P1480: P1481: P1482: P1483: P1484: P1485: P1486: P1487: P1488: P1489: P1490: P1491: P1492: P1493: P1494: P1495: P1496: P1497: P1498: P1499: P1500: P1501: P1502: P1503: P1504: P1505: P1506: P1507: P1508: P1509: P1510: P1511: P1512: P1513: P1514: P1515: P1516: P1517: P1518: P1519: P1520: P1521: P1522: P1523: P1524: P1525: P1526: P1527: P1528: P1529: P1530: P1531: P1532: P1533: P1534: P1535: P1536: P1537: P1538: P1539: P1540: P1541: P1542: P1543: P1544: P1545: P1546: P1547: P1548: P1549: P1550: P1551: P1552: P1553: P1554: P1555: P1556: P1557: P1558: P1559: P1560: P1561: P1562: P1563: P1564: P1565: P1566: P1567: P1568: P1569: P1570: P1571: P1572: P1573: P1574: P1575: P1576: P1577: P1578: P1579: P1580: P1581: P1582: P1583: P1584: P1585: P1586: P1587: P1588: P1589: P1590: P1591: P1592: P1593: P1594: P1595: P1596: P1597: P1598: P1599: P1600: P1601: P1602: P1603: P1604: P1605: P1606: P1607: P1608: P1609: P1610: P1611: P1612: P1613: P1614: P1615: P1616: P1617: P1618: P1619: P1620: P1621: P1622: P1623: P1624: P1625: P1626: P1627: P1628: P1629: P1630: P1631: P1632: P1633: P1634: P1635: P1636: P1637: P1638: P1639: P1640: P1641: P1642: P1643: P1644: P1645: P1646: P1647: P1648: P1649: P1650: P1651: P1652: P1653: P1654: P1655: P1656: P1657: P1658: P1659: P1660: P1661: P1662: P1663: P1664: P1665: P1666: P1667: P1668: P1669: P1670: P1671: P1672: P1673: P1674: P1675: P1676: P1677: P1678: P1679: P1680: P1681: P1682: P1683: P1684: P1685: P1686: P1687: P1688: P1689: P1690: P1691: P1692: P1693: P1694: P1695: P1696: P1697: P1698: P1699: P1700: P1701: P1702: P1703: P1704: P1705: P1706: P1707: P1708: P1709: P1710: P1711: P1712: P1713: P1714: P1715: P1716: P1717: P1718: P1719: P1720: P1721: P1722: P1723: P1724: P1725: P1726: P1727: P1728: P1729: P1730: P1731: P1732: P1733: P1734: P1735: P1736: P1737: P1738: P1739: P1740: P1741: P1742: P1743: P1744: P1745: P1746: P1747: P1748: P1749: P1750: P1751: P1752: P1753: P1754: P1755: P1756: P1757: P1758: P1759: P1760: P1761: P1762: P1763: P1764: P1765: P1766: P1767: P1768: P1769: P1770: P1771: P1772: P1773: P1774: P1775: P1776: P1777: P1778: P1779: P1780: P1781: P1782: P1783: P1784: P1785: P1786: P1787: P1788: P1789: P1790: P1791: P1792: P1793: P1794: P1795: P1796: P1797: P1798: P1799: P1800: P1801: P1802: P1803: P1804: P1805: P1806: P1807: P1808: P1809: P1810: P1811: P1812: P1813: P1814: P1815: P1816: P1817: P1818: P1819: P1820: P1821: P1822: P1823: P1824: P1825: P1826: P1827: P1828: P1829: P1830: P1831: P1832: P1833: P1834: P1835: P1836: P1837: P1838: P1839: P1840: P1841: P1842: P1843: P1844: P1845: P1846: P1847: P1848: P1849: P1850: P1851: P1852: P1853: P1854: P1855: P1856: P1857: P1858: P1859: P1860: P1861: P1862: P1863: P1864: P1865: P1866: P1867: P1868: P1869: P1870: P1871: P1872: P1873: P1874: P1875: P1876: P1877: P1878: P1879: P1880: P1881: P1882: P1883: P1884: P1885: P1886: P1887: P1888: P1889: P1890: P1891: P1892: P1893: P1894: P1895: P1896: P1897: P1898: P1899: P1900: P1901: P1902: P1903: P1904: P1905: P1906: P1907: P1908: P1909: P1910: P1911: P1912: P1913: P1914: P1915: P1916: P1917: P1918: P1919: P1920: P1921: P1922: P1923: P1924: P1925: P1926: P1927: P1928: P1929: P1930: P1931: P1932: P1933: P1934: P1935: P1936: P1937: P1938: P1939: P1940: P1941: P1942: P1943: P1944: P1945: P1946: P1947: P1948: P1949: P1950: P1951: P1952: P1953: P1954: P1955: P1956: P1957: P1958: P1959: P1960: P1961: P1962: P1963: P1964: P1965: P1966: P1967: P1968: P1969: P1970: P1971: P1972: P1973: P1974: P1975: P1976

- QVMe

Init()

Connect()

Disconnect()

SetMeasTrig()

MeasureLADC()

MeasureHADC()

GetMeasResult()

StartFFT()

GetFFTResult()

BlockRead()

• QVMe

Init();

Remarks

初始化 QVMe 通道。执行该语句之后：

- 1) 所有输入输出继电器断开；
- 2) 低速通道和高速通道未设定有效量程并且停止数据转换；
- 3) 低速通道默认 10KHz 滤波；
- 4) 存储在RAM中的数据是最后的测量数据, 无法回读之前测量结果。所有返回值为0.

注意:

使用 QVMe 测试结束时建议调用此函数初始化。

```
qvme0.Init(); // Connect Logic Channel 0 Input Relay
```

- QVMe

Connect(void);

Remarks

接通 QVMe 通道的输入输出继电器。在测量之前应先接通输入输出继电器。

```
qvme0.Connect(); // Connect Logic Channel 0 Input Relay
```

Disconnect(void);

Remarks

断开 QVMe 通道的输入输出继电器。在测量完成之后应断开输入输出继电器。

```
qvme0.Disconnect(); // Connect Logic Channel 0 Input Relay
```

• QVMe

```
SetMeasTrig(double vTrig, TRIG_MODE trigMode, double  
hysteresisValue = 0.0);
```

Remarks

为特定逻辑通道设置触发阈值和触发方向。

本功能主要用于同步测量。

注意:

当 MeasureLADC()或 MeasureHADC()中的 measMode 选择 MEAS_NORMAL 模式时，此函数设置的触发阈值无效。

Parameters

vTrig 触发阈值，范围与选定的测量量程相关。

trigMode 触发方向。

TRIG_FALLING 下降沿触发

TRIG_RISING 上升沿触发

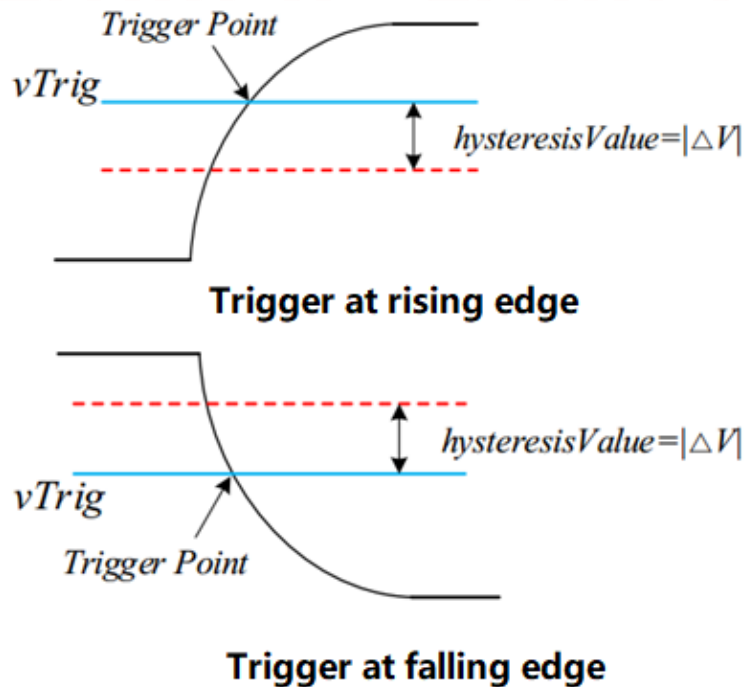
hysteresisValue double 型变量，单位 V。设置触发电压的滞回值。

该参数默认为 0。

• QVMe

注意:

- 1) 滞回参数 *hysteresisValue* 仅可设置为非负值;
- 2) 在上升沿 (TRIG_RISING) 触发的情况下, 当判定到测量结果先小于 ($vTrig - hysteresisValue$) 值之后又大于 $vTrig$ 值时, 此时在 $vTrig$ 处判定为触发;
- 3) 在下降沿 (TRIG_FALLING) 触发的情况下, 当判定到测量结果先大于 ($vTrig + hysteresisValue$) 值之后又小于 $vTrig$ 值时, 此时在 $vTrig$ 处判定为触发。



• QVMe

MeasureLADC(UINT *sampleTimes* = 100, double *sampleInterval* = 10, QVMe_LADC_VRANG *VRange* = QVMe_LADC_10V, QVMe_LADC_FILTER *Filter* = QVMe_LADC_10KHz, FLOATMEASMODE *measMode* = MEAS_NORMAL, UINT *T1* = 0, UINT *T2* = 0, UINT *T3* = 0, UINT *T4* = 0);

Remarks

设置低速通道工作参数并进行采样，将采样结果存储在板上存储器中。
在调用此函数前需调用 **Connect()**函数接通输入输出继电器。

Parameters

sampleTimes 采样次数，取值范围 1~65535，默认为 100。

如果需进行 FFT，采样点数范围为 64~32768。

sampleInterval 采样间隔，取值范围 1~1300，单位为 uS。分辨率为 0.1uS，默认为 10uS。

VRange 测量量程，取值如下：

QVMe_LADC_100V, QVMe_LADC_50V, QVMe_LADC_20V, QVMe_LADC_10V

QVMe_LADC_5V, QVMe_LADC_2V, QVMe_LADC_1V, QVMe_LADC_500MV

QVMe_LADC_200MV, QVMe_LADC_100MV

默认为 QVMe_LADC_10V。

Filter 滤波器设置，取值如下：

QVMe_LADC_400KHz, QVMe_LADC_100KHz

QVMe_LADC_40KHz, QVMe_LADC_10KHz

默认为 QVMe_LADC_10KHz。

• QVMe

Parameters

measMode 测量模式，取值如下：

MEAS_NORMAL: 普通测量模式，在调用函数之后，测量将开始并完成。

MEAS_AWG: AWG 模式，只用于设置。直到调用 **awg** 全局函数，数字转换器才会开始工作。
默认为 **MEAS_NORMAL**。

T1、*T2*、*T3*、*T4*

单位 μS 。分辨率 $10\mu\text{S}$ ，范围：0 ~ ($T1+T2+T3+T4$) ~ 600000。

AWG 测量模式下，设置测量启动时刻。一次 AWG 中，可以启动四次测量。四个参数均可以不写，默认启动一次，启动时刻为 0。

对于这种测量模式，建议跟同步启动测量配合使用。

注意：

(1) *T1*、*T2*、*T3*、*T4* 为相对时间。即 *T2* 的启动时间为相对 *T1* 的启动时间，*T3* 的启动时间为相对 *T2* 的启动时间，*T4* 的启动时间为相对 *T3* 的启动时间。

(2) *T1*、*T2*、*T3*、*T4* 这四个参数仅在 **MEAS_AWG** 模式下起作用。

例 1：低速通道采样 1000 点，采样间隔 $1\mu\text{S}$ ，10V 量程，40KHz 滤波，普通测量模式

```
qvme0.MeasureLADC(1000, 1, QVMe_LADC_10V, QVMe_LADC_40KHz, MEAS_NORMAL, 0,0,0,0);
```

• QVMe

MeasureHADC(UINT *sampleTimes* = 100, double *sampleInterval* = 10, QVMe_HADC_VRANG *VRange* = QVMe_HADC_4V, FLOATMEASMODE *measMode* = MEAS_NORMAL, UINT *T1* = 0, UINT *T2* = 0, UINT *T3* = 0, UINT *T4* = 0);

Remarks

设置高速通道工作参数并进行采样，将采样结果存储在板上存储器中。
在调用此函数前需调用 **Connect()**函数接通输入输出继电器。

Parameters

sampleTimes 采样次数，取值范围 1~65535，默认为 100。
如果需进行 FFT，采样点数范围为 64~32768。

sampleInterval 采样间隔，取值范围 0.1~1，单位为 uS。
分辨率为 0.1uS，默认为 1uS。

VRange 测量量程，取值如下：
QVMe_HADC_4V
QVMe_HADC_2V
QVMe_HADC_1V
默认为 QVMe_HADC_4V。

• QVMe

Parameters

measMode 测量模式，取值如下：

MEAS_NORMAL: 普通测量模式，在调用函数之后，测量将开始并完成。

MEAS_AWG: AWG 模式，只用于设置。直到调用 **awg** 全局函数，数字转换器才会开始工作。

默认为 **MEAS_NORMAL**

T1、*T2*、*T3*、*T4*

单位 μS 。分辨率 $10\mu\text{S}$ ，范围： $0 \sim (T1+T2+T3+T4) \sim 600000$ 。

AWG 测量模式下，设置测量启动时刻。一次 AWG 中，可以启动四次测量。四个参数均可以不写，默认启动一次，启动时刻为 0。

对于这种测量模式，建议跟同步启动测量配合使用。

注意：

(1) *T1*、*T2*、*T3*、*T4* 为相对时间。即 *T2* 的启动时间为相对 *T1* 的启动时间，*T3* 的启动时间为相对 *T2* 的启动时间，*T4* 的启动时间为相对 *T3* 的启动时间。

(2) *T1*、*T2*、*T3*、*T4* 这四个参数仅在 **MEAS_AWG** 模式下起作用。

例 1：高速通道采样 1000 点，采样间隔 $0.1\mu\text{S}$ ，4V 量程，普通测量模式

```
qvme0.MeasureHADC(1000, 1, QVMe_HADC_4V, MEAS_NORMAL, 0,0,0,0);
```

• QVMe

```
GetMeasResult (UINT siteNo,  
int sampleNumber = AVERAGE_RESULT);
```

Remarks

获取单一工位测量结果，若要获取某一工位多个采样点结果，则需做个循环。在调用此函数前需调用 MeasureLADC()或 MeasureHADC()函数进行采样。

Parameters

siteNo 选择测量数据工位。0 代表 Site1，1 代表 Site2.....

sampleNumber 返回结果类型标识：

AVERAGE_RESULT 返回所有采样点的平均值

MAXMIN_RESULT 返回所有采样点的最大值

MIN_RESULT 返回所有采样点的最小值

TRIG_RESULT 返回触发时对应采样点的位置，需配合 SetMeasTrig()使用

RMS_RESULT: 返回所有采样点的有效值

Non-negative value N: 返回的是第 N+1 个采样点的测量值。假设 MeasureLADC或 MeasureHADC 的采样次数为 sam，则 $0 \leq N \leq \text{sam}-1$ 。

默认为 AVERAGE_RESULT。

注意:

当 MeasureLADC()或 MeasureHADC()中的 measMode 选择 MEAS_AWG 模式时，GetMeasResult()中的 *sampleNumber* 设置为 RMS_RESULT 无效。

- QVMe

1. 获取 qvme0 的采样平均值。

```
int SiteID=0;
double result[SITENUM] = {0.0};
qvme0.MeasureLADC(1000,1,QVMe_LADC_10V,QVMe_LADC_40KHz,MEAS_NORMAL,0,0,0,0);
for(SiteID=0; SiteID<SITENUM;SiteID++)
{
    result[SiteID] = qvme0.GetMeasResult(SiteID, AVERAGE_RESULT);
}
```

- QVMe

StartFFT(QVMe_FFT_TYPE *fftType*)

Remarks

启动 FFT，此函数对采样数据进行 FFT，FFT 点数范围：64~32768。

在调用此函数之前需先调用 MeasureLADC()函数或者 MeasureHADC()函数进行采样。

注意：

当 MeasureLADC()或 MeasureHADC()中的 measMode 选择 MEAS_AWG 模式时，StartFFT 函数无效。

Parameters

fftType

FFT 类型选择，有两种：

QVMe_DIRECT_FFT: 直接 FFT

QVMe_BLACKMAN_HARRIS_FFT: 加 BLACKMAN_HARRIS 窗 FFT

• QVMe

GetFFTResult(UINT *siteNo*, long *retResult* = THD_RESULT, BYTE *harmonicType* = TOTAL_HARM, BYTE *harmonicNum* = 5, double *FreqBand* = -1)

Remarks

调用此函数获取与 FFT 相关的测试结果。

注意:

当 MeasureLADC()或 MeasureHADC()中的 measMode 选择 MEAS_AWG 模式, GetFFTResult 函数无效。

Parameters

siteNo UINT 型变量, 选择测量数据工位。0 代表 Site1, 1 代表 Site2.....

retResult 返回结果类型:

THD_RESULT 返回 THD

SNR_RESULT 返回 SNR

SINAD_RESULT 返回 SINAD

Non-negative value N: 返回 $N * ((1/\text{sampleInterval}) / \text{sampleTimes})$ 的频点的电压幅值,
 $0 \leq N \leq \text{sampleTimes} - 1$ 。

默认为 THD_RESULT

harmonicType 谐波类型, 用于 THD 计算。取值如下:

TOTAL_HARM: 总谐波

ODD_HARM: 奇次谐波

EVEN_HARM: 偶次谐波

该参数可以不写, 默认为 TOTAL_HARM。

注意: 当返回结果类型为 SNR_RESULT 或 SINAD_RESULT 时, 此参数无效。

• QVMe

Parameters

harmonicNum 谐波次数，用于 THD 计算，
取值范围 2~50。默认为 5。

注意：

当返回结果类型标识为 SNR_RESULT 或 SINAD_RESULT 时，此参数无效。

FreqBand R频带范围，用于 SNR 和 SINAD 计算指定频带范围内的谐波和噪声，单位为 KHz。
取值范围：0~((1/ sampleInterval)/2)*1000。
默认为-1，取全部谐波和噪声。

注意：

- (1) 当取值不在给定范围时，取全部谐波和噪声。
- (2) 当返回结果类型标识为 THD_RESULT 时，此参数无效。

- QVMe

Ex: 使用 GetFFTResult 直接获取 THD、SNR、SINAD。

```
double THD_result[SITENUM]={0.0};
double SNR_result[SITENUM]={0.0};
double SINAD_result[SITENUM]={0.0};
qvme0.Connect();
delay_ms(1);
qvme0.MeasureLADC(4096,1,QVMe_LADC_10V,QVMe_LADC_400KHz,MEAS_NORMAL,0,0,0,0);
// low speed mode, sample=4096, interval=1uS, 10V vrage, 400KHz filter, normal mode
qvme0.StartFFT(QVMe_DIRECT_FFT);// start FFT directly
int SiteID=0;
for(SiteID=0; SiteID<SITENUM;SiteID++)
{
    THD_result[SiteID]=qvme0.GetFFTResult(SiteID,THD_RESULT,TOTAL_HARM,19,-1);
    // total harmonic, harmonics number is 20
    SNR_result[SiteID]=qvme0.GetFFTResult(SiteID,SNR_RESULT,TOTAL_HARM,5,19);
    SINAD_result[SiteID]=qvme0.GetFFTResult(SiteID,SINAD_RESULT,TOTAL_HARM,5,19);
}
for(SiteID=0; SiteID<SITENUM;SiteID++)
{
    // THD->SetTestResult(SiteID,0, THD_result[SiteID]);
    // SNR->SetTestResult(SiteID,0, SNR_result[SiteID]);
    // SINAD->SetTestResult(SiteID,0, SINAD_result[SiteID]);
}
```

• QVMe

BlockRead(BYTE *siteNo*, long *startAddr*, long *size*, double **buffer*, QVMe_MEAS_RET *retType* = QVMe_SAMPLE_DATA);

Remarks

此函数读取一个数据块，数据块的起始地址和大小由用户定义。

Parameters

siteNo UINT 型变量，选择测量数据工位。0 代表 Site1，1 代表 Site2.....

startAddr 数据块的起始地址，非负整数。

Size 数据块的大小，正整数。

Buffer 存放数据结果的数组，由用户定义，如：double *fft_result*[2048]={0.0};

注意：

(1) *buffer* 数组长度需不小于 *size* 的值，否则可能会发生不可预期的错误；

(2) 建议把该数组定义成全局变量。

retype 返回的数据类型：

QVMe_SAMPLE_DATA: 时域采样数据

QVMe_FFT_DATA: 频域 FFT 数据

需先调用 StartFFT()函数进行 FFT 变换

默认为 QVMe_SAMPLE_DATA。

注意：

当 MeasureLADC()或 MeasureHADC()中的 *measMode* 选择 MEAS_AWG 模式时，BlockRead 中的 *retType* 设置为 QVMe_FFT_DATA 无效。

- QVMe

例 1：读取 FFT 结果

```
int SiteID=0
double fft_result[SITENUM][2048]={0.0};
qvme0.MeasureLADC(2048,5,QVMe_LADC_10V,QVMe_LADC_40KHz,MEAS_NORMAL, 0, 0, 0, 0);
// low speed mode, sample=2048, interval=5uS, 10V vrage, 40KHz filter, normal mode.
qvme0.StartFFT(QVMe_DIRECT_FFT);// satrt FFT directly
for(SiteID=0; SiteID<SITENUM;SiteID++)
{
    qvme0.BlockRead(SiteID, 0, 2048, fft_result[SiteID], QVMe_FFT_DATA);
}
```

例 2：直接读取数据

```
int SiteID=0
double data_result[SITENUM][2048]={0.0};
qvme0.MeasureLADC(2048,5,QVMe_LADC_10V,QVMe_LADC_40KHz,MEAS_NORMAL, 0, 0, 0, 0);
// low speed mode, sample=2048, interval=5uS, 10V vrage, 40KHz filter, normal mode.
for(SiteID=0; SiteID<SITENUM;SiteID++)
{
    qvme0.BlockRead(SiteID, 0, 2048, data_result[SiteID], QVMe_SAMPLE_DATA);
}
```

• QVMe样例

```
DUT_API int test_Average(short funcindex, LPCTSTR funclabel)
{
    //{{AFX_STG_PARAM_PROTOTYPES
    CParam *Average = StsGetParam(funcindex, "Average");
    //}}AFX_STG_PARAM_PROTOTYPES
    double avg_result[SITENUM] = {0.0};
    qvme0.Connect();
    delay_ms(1);

    qvme0.MeasureLADC(200, 5, QVMe_LADC_5V, QVMe_LADC_40KHz, MEAS_NORMAL);
    //sample is 200; interval time is 5us
    for(int SiteID=0; SiteID<SITENUM; SiteID++)
    {
        avg_result[SiteID] = qvme0.GetMeasResult(SiteID, AVERAGE_RESULT);
        // Average->SetTestResult(SiteID, 0, avg_result[SiteID]);
    }

    // TODO: Add your function code here
    return 0;
}
```

• QVMe样例

```
DUT_API int test_Synchronization_with_FPVIe(short funcindex, LPCTSTR funclabel)
{
    //{{AFX_STG_PARAM_PROTOTYPES
    CParam *TirgQVMe = StsGetParam(funcindex,"TirgQVMe");
    CParam *TirgFPVI10 = StsGetParam(funcindex,"TirgFPVI10");
    //}}AFX_STG_PARAM_PROTOTYPES

    int sample = 100;
    double Trig = 5;//trigger level
    double Trig_Point_qvme[SITENUM] = { 0.0 };
    double Trig_Point_fpvi[SITENUM] = { 0.0 };
    double result_qvme[SITENUM] = { 0.0 };
    double result_fpvi[SITENUM] = { 0.0 };
    int sam = 50; //data size is 50
    int interval = 20; //interval time is 20us unit=uS
    double awg_pattern[1000] = { 0.0 };

    qvme0.Connect();
    delay_ms(1);

    STSAWGCreateRampData(&awg_pattern[0], sam, 1, 8, 3);
    //Ramp data,start address =0; data size=50, wavecycle=1, startValue=8V, stopValue=3V

    fpvie0.AwgLoader("Vst", FV, FPVIe_10V, FPVIe_10MA, awg_pattern, sam);
    //load the data of "awg_pattern"(sam=50), named "Vst"

    fpvie0.Set(FV, 8, FPVIe_10V, FPVIe_10MA, FPVIe_RELAY_ON);
    // Call 'Set()' to make sure that the mode and ranges are exactly the same with 'Vst' of its 'AwgLoader()'

    fpvie0.AwgSelect("Vst", 0, sam - 1, sam - 1, interval);
    //Select pattern of 'Vst', start step=0, stop=sam-1; loop= sam-1; interval time=20us

    fpvie0.SetMeasVTrig(Trig, TRIG_FALLING);//set V_trigger level as 5V, falling edge
    fpvie0.MeasureVI(sam, interval, FPVIe_MV_X1, FPVIe_MI_X1, MEAS_AWG);
}
```

• QVMe样例

```
qyme0.MeasureLADC(100, 10, QVme_LADC_10V, QVme_LADC_10KHz, MEAS_AWG);
qyme0.SetMeasTrig(Trig, TRIG_FALLING); //set V_trigger level as 5V, falling edge
//Low speed mode, SampleTimes=100, samplePeriod=10US, AWG Mode

STSEnableAWG(&fpvie0);           // fpvie0 enable AWG
STSEnableMeas(&fpvie0);          // fpvie0 enable measuring
qyme0.EnbGlobalStartMeas();       // qyme0 enable measuring
qyme0.GlobalStart();

STSAWGRun(); //Enable AWG and measurement synchronously

int SiteID = 0;
for (SiteID = 0; SiteID < SITENUM; SiteID++)
{
    Trig_Point_qyme[SiteID] = qyme0.GetMeasResult(SiteID, TRIG_RESULT);
    //qyme0 Get TRIG_RESULT
    result_qyme[SiteID] = qyme0.GetMeasResult(SiteID, Trig_Point_qyme[SiteID]);
    //qyme0 Get Value of TRIG_RESULT
    Trig_Point_fpvi[SiteID] = fpvie0.GetMeasResult(SiteID, MVRET, TRIG_RESULT);
    //fpvie0 Get TRIG_RESULT
    result_fpvi[SiteID] = fpvie0.GetMeasResult(SiteID, MVRET, Trig_Point_fpvi[SiteID]);
    //fpvie0 Get Value of TRIG_RESULT

    //TingQVMe->SetTestResult(SiteID, 0, result_qyme[SiteID]);
    //TrigFPVI10->SetTestResult(SiteID, 0, result_fpvi[SiteID]);
}

// TODO: Add your function code here
return 0;
}
```



Make Testing More Valuable